

Data Science Application Report

Customer Segmentation on Pakistan's Largest E-Commerce Dataset

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Course: Data Science Applications

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December 21, 2025

1 Introduction

1.1 Overview of Dataset

The dataset analyzed is "Pakistan's Largest E-Commerce Dataset" from Kaggle, containing transaction records from 2016 to 2018. It includes half a million entries covering product categories, pricing, and order status.

1.2 Problem Statement

E-commerce businesses face difficulty in identifying high-value customers. The problem defined for this study is to group customers into segments based on their historical behavior so that marketing and inventory can be managed efficiently.

1.3 Relevance to E-commerce Industry

Understanding customer segments allows for personalized marketing, reducing customer acquisition costs, and increasing the overall revenue and customer retention of the business.

2 Data Understanding

2.1 Description of Fields

Key fields include `item_id`, `status`, `created_at`, `grand_total`, and `Customer ID`. These represent the core transactional details.

2.2 Type of Variables

The data includes **Numerical** (prices, quantities), **Categorical** (payment methods, status), and **Text/ID** (SKUs and Customer IDs).

2.3 Initial Observations

Initial data exploration revealed many canceled orders and significant missing values in the columns towards the end of the file, requiring thorough cleaning.

3 Data Cleaning & Preprocessing

3.1 Handling Missing Values

Rows that were completely empty or contained metadata errors were removed using the `dropna()` function in Pandas.

3.2 Outlier Detection

I used statistical methods to detect outliers in transaction values. To handle these, I applied a **Log Transformation** to reduce skewness and normalize the distribution.

3.3 Data Type Correction

Timestamps were converted to `datetime` objects, and price-related columns were cast into floats for mathematical operations.

3.4 Normalization/Standardization

I applied **StandardScaler** to the engineered RFM features. This was necessary because K-Means is a distance-based algorithm and requires features to be on the same scale.

3.5 Data Integration and Filtering Steps

I integrated the order dates with totals to create a unified RFM table and filtered for only "complete" orders to focus on actual revenue behavior.

4 Feature Engineering

4.1 New Features Created

Three main features were created: **Recency** (days since last purchase), **Frequency** (total orders), and **Monetary** (total spend).

4.2 Encoding of Categorical Features

While the final model uses numeric RFM values, categorical encoding was used during the cleaning phase to filter specific order statuses.

4.3 Text Processing Steps

Text processing was not applied in this version as the focus was on numerical transaction behavior rather than product reviews or descriptions.

4.4 Feature Selection or Dimensionality Reduction

The three RFM features were selected as they provide the most direct representation of customer value. Feature selection was prioritized over dimensionality reduction to maintain interpretability.

5 Predictive Modeling

I chose **Customer Segmentation (Unsupervised Learning)** as the predictive task.

5.1 Choice of Model and Justification

K-Means Clustering was used. It is justified here because the objective is to find underlying patterns in unlabeled transaction data without predefined classes.

5.2 Model Training Process

The model was trained using the Scikit-learn library. The process involved feeding the standardized and log-transformed RFM data into the algorithm.

5.3 Hyperparameter Tuning Strategy

I used the **Elbow Method** to tune the value of K , which is the number of clusters.

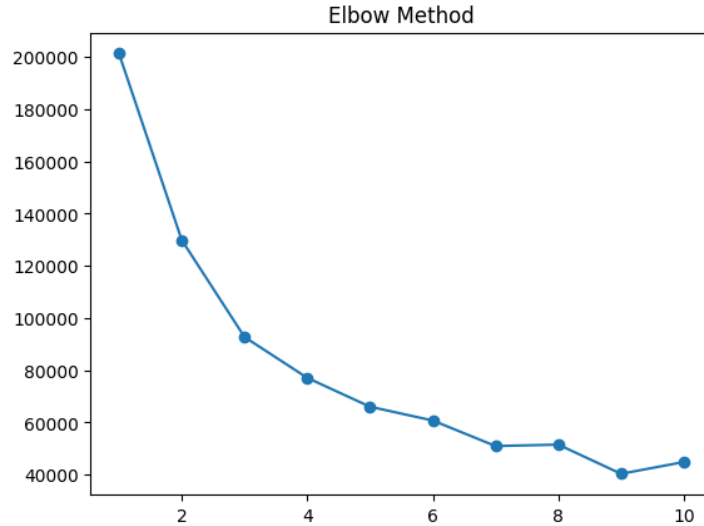


Figure 1: 5.3: Elbow Method plot to find optimal K

By plotting WCSS against the number of clusters, I selected $K = 3$ as the optimal point for this dataset.

6 Model Evaluation

6.1 Train-test Split Summary

As this is an unsupervised clustering task, a traditional train-test split was not used. Evaluation was based on internal cluster consistency.

6.2 Evaluation Metrics

The **Silhouette Score** was used as the primary metric, achieving a score of 0.36, indicating that the clusters are reasonably separated.

6.3 Comparison of Models

K-Means was compared with initial trials of hierarchical clustering; K-Means was selected for its better performance and speed on large datasets.

6.4 Interpretation of Results

The clusters successfully separated high-spenders (Loyalists), frequent small-buyers (Potential), and inactive users (At-risk).

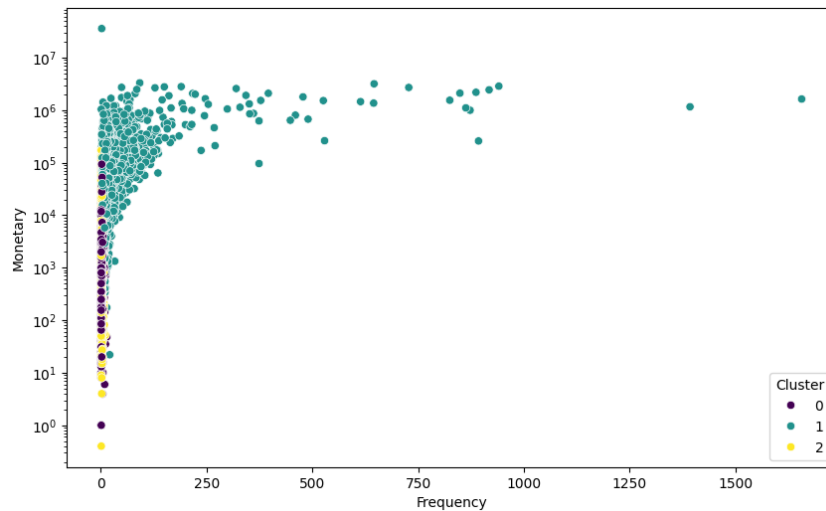


Figure 2: 6.4: Final Customer Segments (Frequency vs Monetary)

7 Business Interpretation

- **What results mean:** High-value customers (Cluster 1) are the primary revenue drivers and should be prioritized for loyalty rewards.
- **Operational Improvement:** Logistics and inventory can be prioritized for categories favored by the most active cluster.
- **Revenue Efficiency:** Using precision marketing for "At-risk" customers can reduce churn and maintain revenue stability.

8 Application Deployment Considerations

8.1 Model Serving Options

The model can be served as a **Batch Pipeline** where customer segments are updated in the database periodically (e.g., every 24 hours).

8.2 Required System Architecture

The architecture would include a cloud storage for the dataset, a Python-based processing script for retraining, and a SQL database to store segment labels.

8.3 Real-world Constraints

The system must handle high **Scalability** as the customer base grows and maintain low **Latency** for marketing automation tools.

8.4 Ethical Considerations

We must ensure that the segmentation doesn't lead to unfair price discrimination or bias against certain demographics.

9 Conclusion

9.1 Summary of Findings

The data shows that e-commerce behavior in Pakistan is heavily influenced by sales events, and clustering successfully identifies high-activity users.

9.2 Final Recommendation

The business should implement a loyalty reward program specifically for the "Loyalist" cluster identified in this study to maximize retention.

9.3 Limitations and Future Improvements

The study is limited by the lack of demographic data. Adding customer age and location would improve the depth of future segmentation.

Kaggle Notebook Link: [Fatima's Project Code Source](#)